

Memory3: Language Modeling with Explicit Memory

Authors: Hongkang Yang, Zehao Lin, Wenjin Wang, Hao Wu, Zhiyu Li, Bo Tang, Wenqiang Wei, Jinbo Wang, Zeyun Tang, Shichao Song, Chenyang Xi, Yu Yu, Kai Chen, Feiyu Xiong, Linpeng Tang, and Weinan E([鄂维南](#))—1 Jul 2024

Institution:

1. Center for LLM, Institute for Advanced Algorithms Research, Shanghai
2. Moqi Inc
3. Center for Machine Learning Research, Peking University

Memory³: Language Modeling with Explicit Memory

Hongkang Yang¹, Zehao Lin¹, Wenjin Wang¹, Hao Wu¹, Zhiyu Li¹, Bo Tang¹, Wenqiang Wei¹, Jinbo Wang¹, Zeyun Tang¹, Shichao Song¹, Chenyang Xi¹, Yu Yu¹, Kai Chen¹,
Feiyu Xiong¹, Linpeng Tang², and Weinan E^{*3,1}

¹Center for LLM, Institute for Advanced Algorithms Research, Shanghai

²Moqi Inc

³Center for Machine Learning Research, Peking University

July 1, 2024

* Also at School of Mathematical Sciences, Peking University and AI for Science Institute
Corresponding authors: xiongyf@iaar.ac.cn, linpengt@myscale.com, weinan@math.pku.edu.cn

Motivation

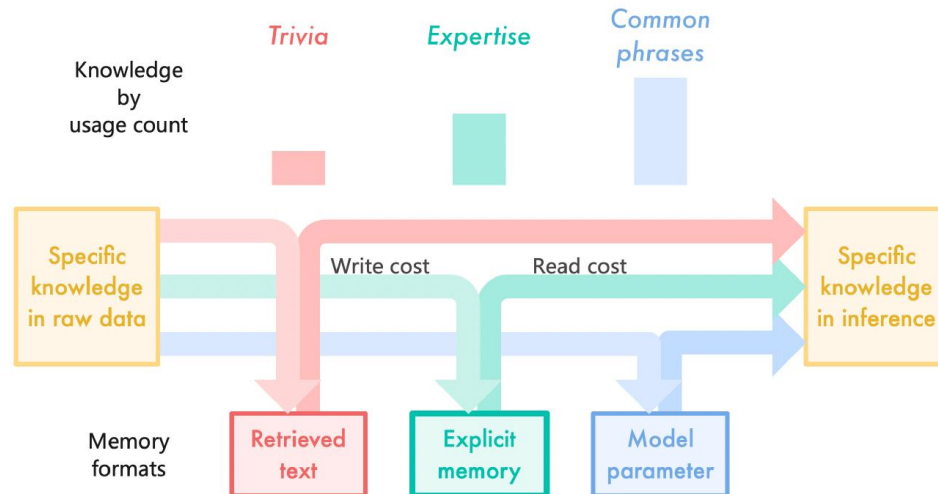
- Externalize **some knowledge** from model parameters into **explicit memory**
 - to reduce the number of model parameters and training costs.
- Design an **efficient explicit memory read and write mechanism** to dynamically call the required knowledge during inference

What is Knowledge?

- computation graph
- Homomorphism
- Circuit
- Deepseek解释:
 - 在大语言模型（LLM）内部，“知识”并不是一段存储好的文本或一个事实，而是一种可复用的、功能特定的“计算电路”或“算法模式”。
 - 当模型处理文本时，它并不是在死记硬背，而是在激活和使用无数个这样的微型“电路”来理解和生成文本。每一种“知识”就是一类功能相同的“电路”的集合。

What knowledge can be separated from the model parameters

- 1. LLM does not possess this knowledge
- 2. Imitable
 - There exists a text $\mathbf{t}_*$ such that for any realization $\mathbf{t}$ of $\mathcal{K}$, the model M using $\mathbf{t}_*$ as prefix can generate each token of the associated output $\mathbf{t}_{\text{out}}$ with high probability, e.g. $\mathbb{P}_M(t_i | \mathbf{t}_* t_0 \dots t_{i-1}) \geq 0.9$ for every $t_i \in \mathbf{t}_{\text{out}}$.
- Claim: Every specific knowledge K is imitable and thus is separable



Imbalanced use of knowledge!!!

Figure 8: Different memory formats with different balances of write cost and read cost. The specific knowledges with high to low usage counts are exemplified by common expressions, expertise and trivia, and are assigned to implicit memory, explicit memory and external information.

Memory Circuitry Theory

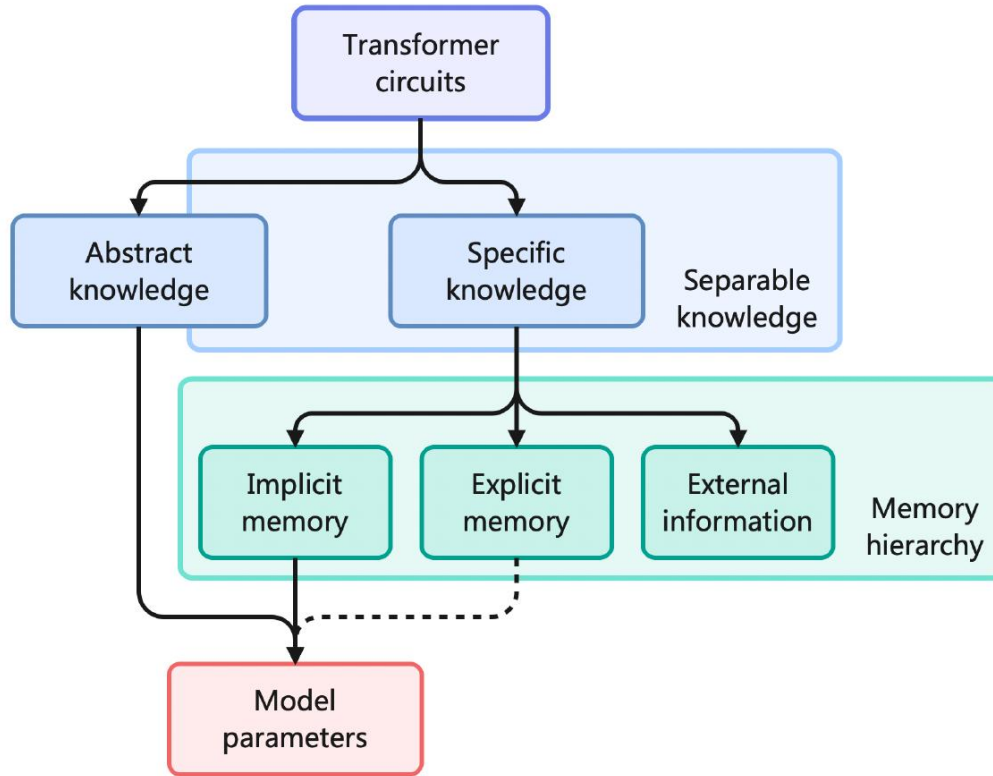


Figure 5: Categorization of knowledge and memory formats. The explicit memories, extracted from model activations, lie half-way between raw data and model parameters, so we use a dotted line to indicate that they may or may not be regarded as parameters.

They want to encourage the LLM to learn only abstract knowledges, with the specific knowledges mostly externalized to the explicit memory bank.

Implicit memory (model parameters): High write cost, low read cost, suitable for storing frequently used knowledge.

Explicit memory: Moderate write and read costs, suitable for storing moderately used knowledge.

External information: Low write cost, high read cost, suitable for storing rarely used knowledge.

Design

- Goal:
 - Moderately low write cost and read cost.
 - Limit the modification to the Transformer architecture to be as little as possible, adding no new trainable parameters
- Write cost: **Key-value vectors of the self-attention** layers to an explicit memory
- Read cost: explicit memories are retrieved from drives **and read by self-attention alongside the usual context key-values.**

Explicit Memory

- Details
 - Numbers: 1.1×10^8 text chunks with length bounded by 128 tokens
 - Shape:
 - (memory layers, 2, key-value heads, sparse tokens, head dimension) = (22, 2, 8, 8, 80)
 - How to retrieve explicit memory:
 - Embedded by BGE-M3
 - Feature of the $\langle \text{cls} \rangle$ token.
 - The vector index is built with FAISS

The decoding process of Memory³ with memory recall

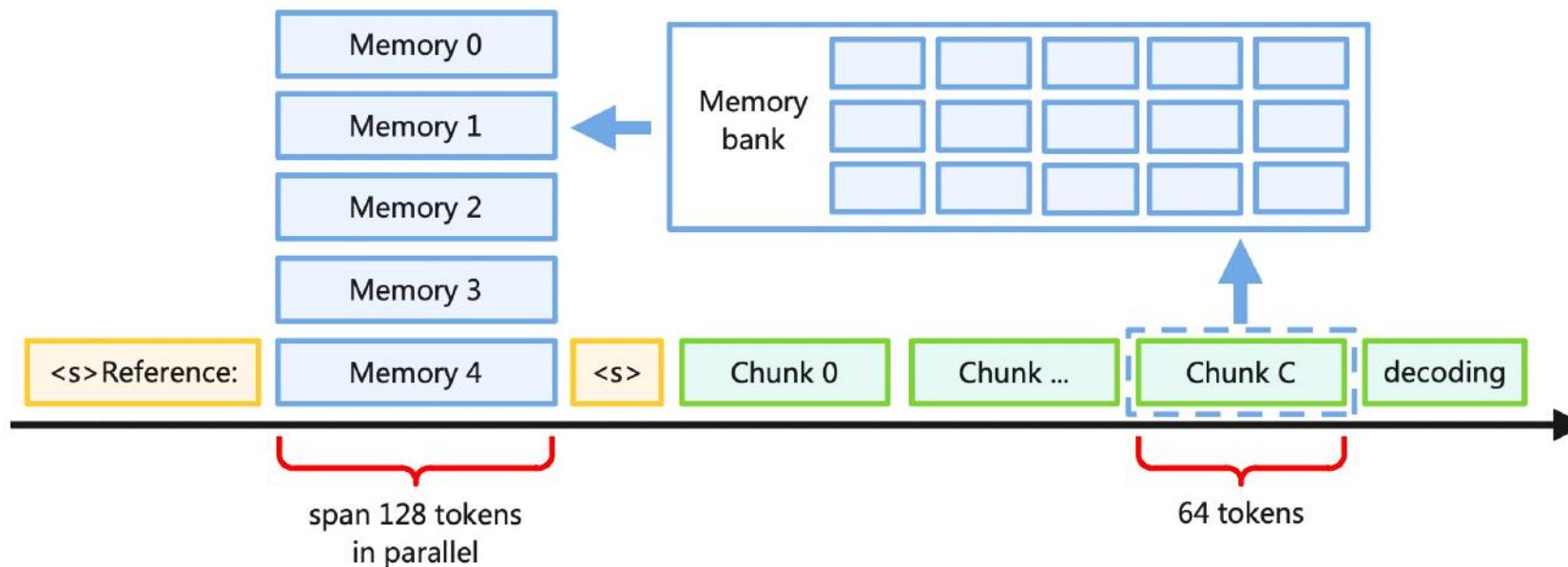


Figure 9: The decoding process of Memory³ with memory recall. Each chunk is a fixed-length interval of tokens, which may belong to either the prompt or generated text.

Methods

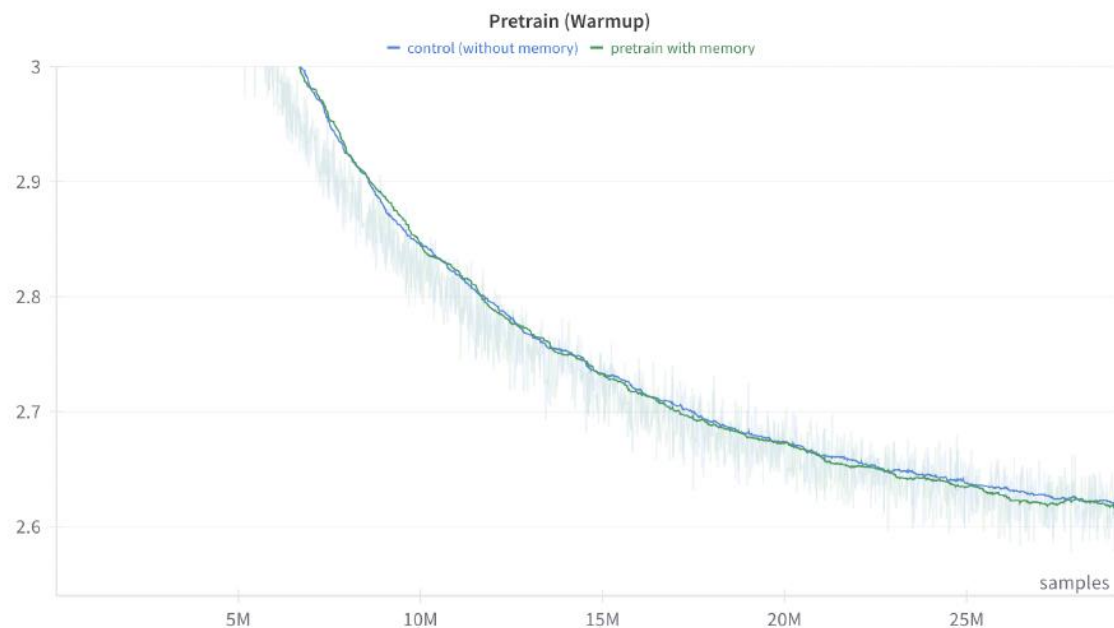
- “ $\langle s \rangle$ Reference:”
 - to help the LLM distinguish between encoding normal texts and encoding references
- Parallel position encoding for all explicit memories
 - The token sparsification is applied after RoPE processes the attention keys,
 - The selected tokens retain their relative positions in the references.

Memory Sparsification and Storage

- (memory layers, 2, key-value heads, sparse tokens, head dimension)
 - Layers: attention to the distant context usually takes place in the middle
 - 25%-75% (44→22)
 - Heads: Use a grouped query attention (GQA) to reduce the amount.
 - 20%(40→8)
 - Tokens (128→8):
 - For each head h at layer l , the selection uses top-8 over the attention weight.
 - Head dimension:
 - Vector quantizer to compress each of the key and value vectors using residual quantizations (残差量化) built with FAISS
 - 7340TB→4.2TB

Training Designs

- Two-stage Pretrain
 - Warmup
 - Continual train
 - The continual train stage involves explicit memories



Pretraining Data

- Pretraining Data
 - 4T token pretrain dataset
 - Deduplication, rule-based filtering, and model-based filtering
- Knowledge Base for Explicit Memories
 - A small subset of our pretrain dataset as the knowledge base.
 - English Wikipedia, WikiHow, the
 - Chinese baike dataset
 - Numbers: 1.1×10^8

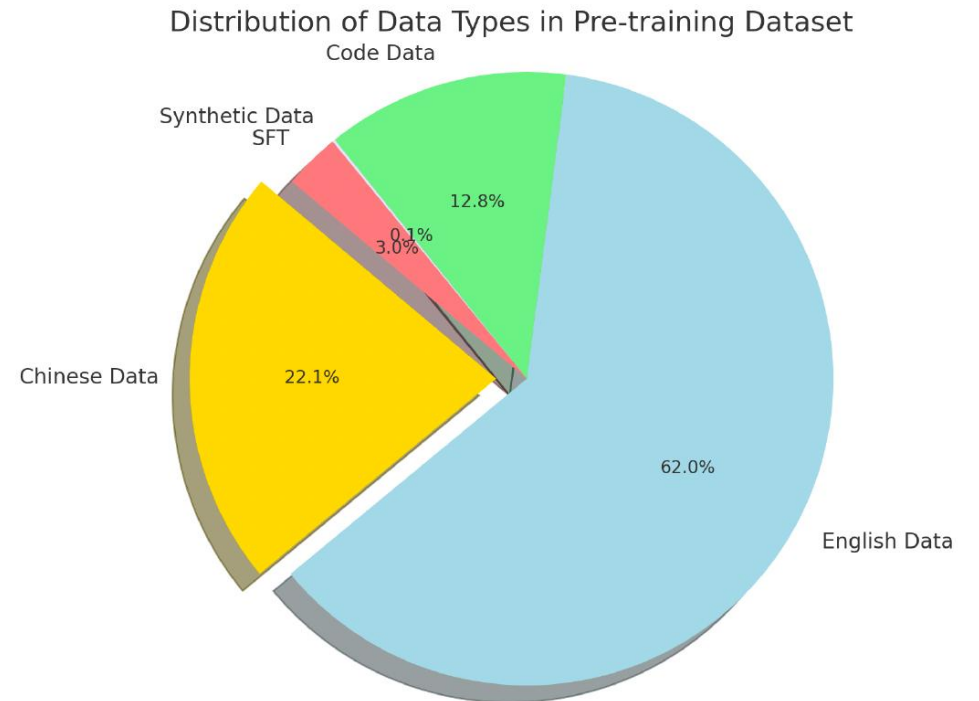


Figure 12: Composition of our pretraining dataset.

Fine-tuning and Alignment

- Supervised Finetuning

Dataset	Source	Number of Samples
UltraChat	HuggingFaceH4/ultrachat_200k	194409
WizardLM	WizardLM/WizardLM_evol_instruct_V2_196k	80662
SlimOrca	Open-Orca/SlimOrca-Dedup	143789
ShareGPT	openchat/openchat_sharegpt4_dataset	3509
Capybara	LDJnr/Capybara	7291
Deita	hkust-nlp/deita-10k-v0	2860
MetaMathQA	meta-math/MetaMathQA	394418
Multi-round Chat	synthetic	20000
Mathematics	synthetic	20000
Commonsense	synthetic	150000
Knowledge	synthetic	270000

Table 15: Composition of SFT dataset.

- Direct Preference Optimization: conversations
 - Conversations; math questions; code questions

Evaluation on General Abilities

LLM	Size	Avg.	English					Chinese	
			ARC-C	HellaSwag	MMLU	Winogrand	GSM8k	CEVAL	CMMLU
Falcon-40B	41B	55.75	61.86	85.28	56.89	81.29	21.46	41.38	42.07
Llama2-7B-Chat	6.5B	46.87	52.90	78.55	48.32	71.74	7.35	34.84	34.40
Llama2-13B-Chat	13B	51.78	59.04	81.94	54.64	74.51	15.24	38.63	38.43
Llama3-8B-it	7.0B	65.77	62.03	78.89	65.69	75.77	75.82	50.52	51.70
Vicuna-13B-v1.5	13B	52.02	57.08	81.24	56.67	74.66	11.30	41.68	41.53
Mistral-7B-v0.1	7.0B	59.15	59.98	83.31	64.16	78.37	37.83	45.91	44.49
Gemma-2B-it	2.0B	36.64	38.02	40.36	55.74	35.29	55.88	8.26	29.94
Gemma-7B-it	7.8B	47.23	51.45	71.96	53.52	67.96	32.22	27.93	25.70
MiniCPM-2B-SFT	2.4B	54.37	47.53	71.95	51.32	67.72	45.26	48.07	48.76
Phi-2	2.5B	55.70	61.09	75.11	58.11	74.35	54.81	34.40	32.04
ChatGLM3-6B	5.7B	54.62	41.38	66.98	50.54	64.25	51.25	54.01	53.91
Baichuan2-7B-Chat	6.5B	55.16	52.73	74.06	52.77	69.77	28.28	53.12	55.38
Qwen1.5-1.8B-Chat	1.2B	49.67	38.74	60.02	45.87	59.67	33.59	55.57	54.22
Qwen1.5-4B-Chat	3.2B	58.15	43.26	69.73	55.55	64.96	52.24	61.89	59.39
Qwen1.5-7B-Chat	6.5B	64.80	56.48	79.02	60.52	66.38	54.36	68.20	68.67
Memory ³ -SFT	2.4B	63.31	58.11	80.51	59.68	74.51	52.84	59.29	58.24
with vector compression	2.4B	63.33	57.94	80.65	59.66	75.14	52.24	59.66	58.05
without memory	2.4B	60.80	57.42	73.14	57.29	74.35	51.33	56.32	55.72

LLM	Avg.	English					Chinese	
		ARC-C	HellaSwag	MMLU	Winogrand	GSM8k	CEVAL	CMMLU
Warmup	42.13	40.27	64.57	41.62	61.96	5.23	40.12	41.17
Continual train	45.12	42.66	79.21	41.81	59.43	6.29	42.20	44.21
- without memory	42.89	42.15	66.98	39.79	61.80	6.44	39.97	43.13
SFT	63.31	58.11	80.51	59.68	74.51	52.84	59.29	58.24
- without memory	60.80	57.42	73.14	57.29	74.35	51.33	56.32	55.72

Table 22: Performance of Memory³-2B at different stages of training. The setup of the evaluation tasks is the same as in Table 16.

Evaluation of Hallucination

LLM	Size	Avg.	English				Chinese
			HaluE-QA	HaluE-Dialogue	TruQA-MC1	TruQA-MC2	HalluQA
Falcon-40B	41B	35.37	46.84	40.80	27.29	41.71	20.18
Llama2-13B	13B	28.01	23.34	31.05	25.95	36.89	22.81
Vicuna-13B-v1.5	13B	37.07	24.93	37.35	35.13	50.88	N/A
Baichuan2-13B	13B	37.64	46.02	45.45	26.81	39.79	30.12
Gemma-7B	7.8B	37.03	50.91	48.19	20.69	46.65	18.71
Mistral-7B-v0.1	7.0B	34.18	40.68	37.64	28.03	42.60	21.93
Llama2-7B	6.5B	36.80	52.46	51.93	25.09	38.94	15.59
Baichuan2-7B	6.5B	38.63	62.33	47.84	23.01	37.46	22.51
ChatGLM3-6B	5.7B	40.96	43.38	50.03	33.17	49.87	28.36
Qwen1.5-4B-Chat	3.2B	33.30	24.64	37.72	29.38	44.74	30.00
Phi-2	2.5B	38.31	50.71	39.55	31.09	44.32	25.89
MiniCPM-SFT	2.4B	36.47	49.24	47.80	24.11	37.51	23.71
Gemma-2B	2.0B	38.04	53.41	52.22	24.60	39.78	20.18
Qwen1.5-1.8B-Chat	1.2B	37.52	47.18	52.11	26.68	40.57	21.05
Memory ³ -SFT	2.4B	48.60	56.61	53.91	38.80	57.72	35.96

Table 19: Evaluation of hallucination. HaluE and TruQA denote HaluEval and TruthfulQA, respectively. Bolded numbers are the best results. The model sizes only include non-embedding parameters. Vicuna-13B-v1.5 gets one N/A since that entry is near zero and seems abnormal.

Naïve RAG vs M3

- All use 5-shot prompting.
- Retrieval from the same FAISS indices.
 - Naïve RAG: text injection
 - M3: Explicit memories injection

LLM	JEC-QA			MED		
	3 refs	5 refs	7 refs	3 refs	5 refs	7 refs
Memory ³ -2B-SFT		39.38			56.22	
MiniCPM-2B-SFT	38.83	37.65	37.94	53.73	53.29	52.84
Gemma-2B	28.16	28.06	25.29	42.04	42.49	42.96
Gemma-2B-it	30.04	31.13	29.34	41.70	43.24	42.66
Llama-2-7B	28.06	24.70	24.90	45.14	44.43	37.96
Llama-2-7B-Chat	26.18	25.10	25.20	48.18	47.29	39.39
Phi-2	25.00	25.30	23.32	50.05	45.42	45.59
Qwen1.5-1.8B-Chat	42.98	43.87	41.50	52.16	52.50	52.16
Qwen1.5-4B-Chat	51.98	50.49	50.99	61.19	61.02	61.06

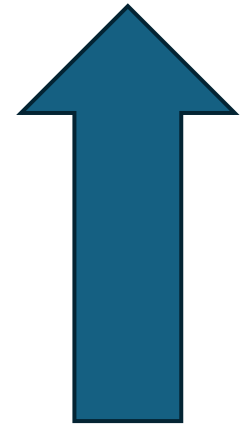
Table 20: Comparison with RAG on professional tasks.

Inference Speed

- Decoding speed on an A800 GPU
- Setting is same to Table 20

LLM	Size	Local server		End-side device	
		with retrieval	w/o retrieval	with retrieval	w/o retrieval
Memory ³ -2B	2.4B	733.0	1131	27.6	44.36
MiniCPM-2B	2.4B	501.5	974.0	21.7	51.79
Gemma-2B-it	2.0B	1581	2056	22.0	29.23
Gemma-7B-it	7.8B	395.6	1008	9.5	18.61
Mistral-7B-Instruct-v0.1	7.0B	392.9	894.5	11.1	28.7
Llama-2-7B-Chat	6.5B	382.8	1005	10.0	23.19
Llama-2-13B-Chat	13B	241.1	632.5	2.5	5.44
Qwen1.5-1.8B-Chat	1.2B	908.2	1770	-	-
Qwen1.5-4B-Chat	3.2B	460.7	1002	22.3	53.39
Qwen1.5-7B-Chat	6.5B	365.8	894.5	-	-
Phi-2	2.5B	622.2	1544	-	-

Table 21: Inference throughput, measured by tokens per second.



Conclusion

- M3 incorporates **explicit memory as a third form of memory**, alongside implicit memory (parameters) and working memory (context key-values), and **explanation from memory circuitry theory**.
- M3 **is trained from pre-training** and evaluated on general performance, hallucination, inference cost, and so on.